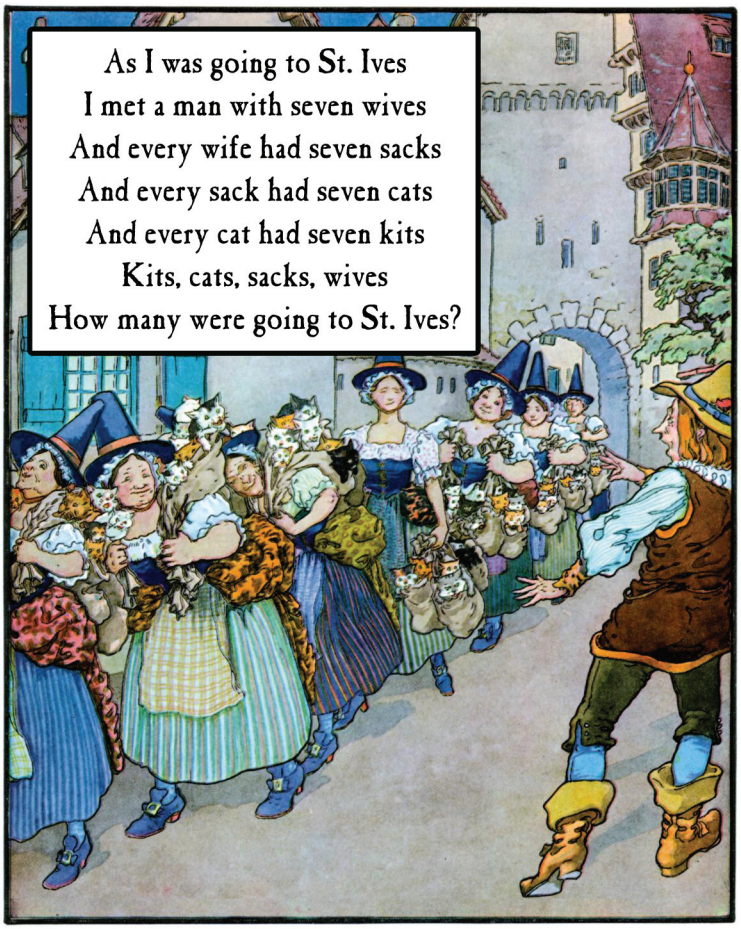
Pre-Activity Preparation

Interpreting Word Problems

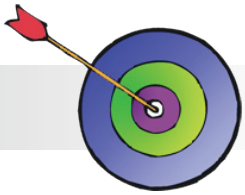


Do you remember this
old nursery rhyme riddle?

Ask this riddle of any second grader and be sure that he or she will furiously begin to multiply seven by seven by seven...

After all the multiplying and adding, the second grader is usually dismayed to find that only **one** was going ***to*** St Ives.

Word problems, though often tricky, are not meant to be riddles. One of the goals of math class is to use the skills learned in the course to solve problems that are presented verbally rather than symbolically. This activity highlights the process for translating from English to mathematical symbols.



Learning Objectives

- Translate key English words to symbolic notation
- Translate a symbolic presentation into English
- Use a symbolic representation to define a problem statement



Terminology

Previously Used

even
odd
operation
proportion
ratio
simplify

New Terms to Learn

See the key word chart on the following pages.

Building Mathematical Language



Working a word problem consists of two parts—*translating* English words and their underlying meaning into a mathematical expression or equation, and then **simplifying** the expression or *solving* the equation.

Students are commonly taught to look for *key words* or phrases when working word problems. The process is to translate each key word or phrase into its corresponding math **operation** symbol and to work with whatever expression or equation results. The key words usually translate to operations on numbers or phrases.

Unfortunately, the key word method does not always take into account the meaning or context of the problem. Scanning the problem for numbers and key words alone may miss the whole point. In the riddle, for example, the man was going *to* St. Ives; nowhere does it state that those whom he met were also going to St. Ives. None of the math words were relevant to the actual riddle. **

****Historical note**: This is a very old riddle; notes about it can be found on the web. (Try searching by the first line of the poem.) Most accounts of the modern wording date to the 17th century, but the first account of solving the mathematics ($7 \times 7 \times 7 \times 7$) dates to about 1600 BC.

Key Words	English Phrase	Symbolic Phrase
For addition		
altogether	If there are four boys and seven girls, how many are there ***altogether***?	$4 + 7$ *or* $7 + 4$
together	Joe has some dollars and Joan has \$7; ***together***, they have what amount?	$\$x + \$7$ *or* $\$7 + \$x$
in all	Joe has \$4 and Joan has \$7, what do they have ***in all***?	$\$4 + \$7$ *or* $\$7 + \$4$
more than	seven ***more than*** four	$4 + 7$
sum of	the ***sum of*** a number and seven	$x + 7$ *or* $7 + x$
total	What is the ***total*** of four and some number?	$4 + x$ *or* $x + 4$
comparatives ***(phrases such as taller than, bigger than, faster than, etc.)***	If Jan is 64 inches tall and John is 6 inches ***taller than*** Jan, how tall is he?	$64'' + 6''$
	Note that it is not always immediately obvious whether a comparative phrase indicates addition or subtraction; be sure to check the context of the problem.	
For subtraction		
difference	Find the ***difference*** of 4 and 7. / Find the ***difference*** of 7 and 4. (Notice which number comes first!)	$4 - 7$ / $7 - 4$
fewer than	four ***fewer than*** some number	$x - 4$
how many more	***How many more*** is seven than four?	$7 - 4$
less than	four ***less than*** some number	$x - 4$

For subtraction (continued)		
left/left over	What is ***left*** if I take 4 from some number ?	$x - 4$
minus	seven ***minus*** four	$7 - 4$
remains	What ***remains*** if I take some number from 7?	$7 - x$
decreased by	seven decreased by four	$7 - 4$
comparatives ***(phrases such as shorter than, smaller than, slower than, etc.)***	If Pete drives at 80 miles per hour and Steve drives 15 miles per hour ***slower than*** Pete, how fast does Steve drive?	80mph – 15mph
	Note that it is not always immediately obvious whether a comparative phrase indicates addition or subtraction; be sure to check the context of the problem.	
For multiplication		
times	four ***times*** some number	$4x$
every	four in ***every*** row of seven	4×7
at the rate of	***at the rate of*** 4 for every number	$4x$
each	Everyone got four ***each.***	$4x$
product	the ***product*** of seven and a number	$7x$
twice/double	***twice*** a number; ***double*** a number	$2x$
For division		
quotient	Find the ***quotient*** of 7 and 4. → $7 \div 4$ Find the ***quotient*** of 4 and 7. → $4 \div 7$ Notice which number comes first!	
equal pieces	a number is cut into four ***equal pieces***	$x \div 4$
split	a number is ***split*** into four equal pieces	$x \div 4$
average	Find the ***average*** of four numbers whose sum is known.	$x \div 4$
half	Find one ***half*** of 18.	$18 \div 2$
shared equally	18 pieces of candy are ***shared equally*** among 3 people.	$18 \div 3$
divided equally	A 12″ sub sandwich is ***divided equally*** among 4 people.	$12'' \div 4$
Other key words	**If the problem says:**	**Represent the numbers with**
two consecutive numbers	Find two ***consecutive*** numbers...	x and $x + 1$
consecutive odd	Find two ***consecutive odd*** numbers...	x and $x + 2$
consecutive even	Find two ***consecutive even*** numbers...	x and $x + 2$
the sum of two numbers	The ***sum of two numbers*** is 7.	x and $7 - x$

Some statements may sound very similar, but convey very different meanings. For instance:

"The sum of 4 times a number and 5"	"4 times the sum of a number and 5"
Sum signifies addition and *times* signifies multiplication. To understand what we are multiplying by 4, underline it.	
The sum of 4 <u>times a number</u> and 5	4 <u>times the sum</u> of a number and 5
$4 \bullet n + 5$	$4 \bullet (n + 5)$
	Parentheses are needed to show that the key word "times" refers to the sum, not just the number.

Punctuation can clarify the meaning of a phrase in a world problem. "Three minus a number plus seven" can be interpreted in two different ways:

$$(3 - x) + 7 \quad \textbf{or} \quad 3 - (x + 7)$$

If we let $x = 2$, we see that these two interpretations give very different results:

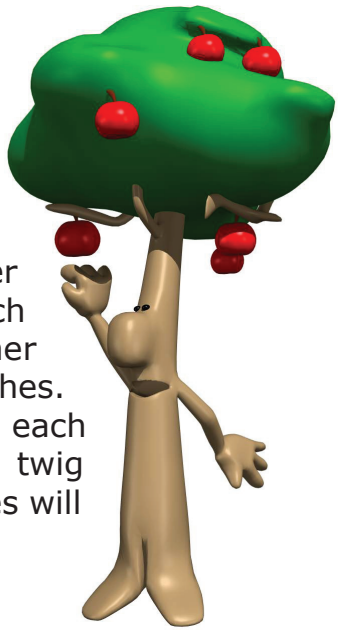
$$(3 - \mathbf{2}) + 7 = 8 \quad \textbf{or} \quad 3 - (\mathbf{2} + 7) = -6$$

If a comma is inserted in the phrase, the correct interpretation is clear:

"Three minus a number, plus seven" can ***only*** mean $(3 - x) + 7$.

Another Classic Riddle

A farmer in California owns a beautiful pear tree. He supplies the fruit to a nearby grocery store. The store owner has called the farmer to see how much fruit is available for purchase. The farmer knows that the main trunk has 24 branches. Each branch has exactly 12 boughs and each bough has exactly 6 twigs. Since each twig bears one piece of fruit, how many apples will the farmer be able to deliver?



Methodologies



The following methodology is useful in *translating* from verbal language to symbolic language. It is a methodology that can be extended to *solving* word problems after you learn to solve equations in the next chapter.

Interpreting a Word Problem

Write an expression for each example.

Example 1: the difference of three times a number and nine

Example 2: the sum of five times a number and seven

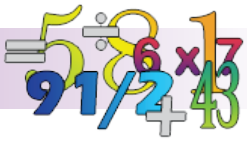
Try It!

Steps in the Methodology		Example 1	Example 2
Step 1 **Read and restate**	Obtain an overview of the problem situation to visualize and obtain general understanding of what is going on. Restate the problem in your own words.	This is a subtraction problem with a variable ("a number")	
Step 2 **Information**	Parse the statement word by word to pull out information in order to produce a list of key words.	1. difference (of __ and __) 2. three 3. times 4. number 5. nine	
Step 3 **Relevancy**	Determine which of the items listed are relevant to the problem situation. Information is relevant when it affects the answer. *We could add "on Wednesdays" to the problem statement and while it would be additional information, it would not be relevant information.*	all five items are relevant	
Step 4 **Target solution**	Articulate what solution is being sought.	an expression for the difference between two quantities	
Step 5 **Translation**	Convert the relevant information in Step 3 to symbolic notation.	1. – (subtraction sign) 2. 3 3. • (multiplication sign) 4. x ("a number") 5. 9	
Step 6 **Model**	Construct the symbolic model that represents the problem.	Answer: $\mathbf{3x - 9}$	

continued on following page

Steps in the Methodology		Example 1	Example 2
Step 7 **Validate**	Validating translations from word problems or phrases to symbolic models can be tricky. One method is to take the symbolic model and translate it back into English. Compare the meaning of this phrase with the original. It is a good idea to check yourself as you work through the problems, especially as you identify the information offered in the problem and translate that information into symbolic and mathematical notation.	We've written an expression ✓ $3x$ | – | 9 three times a number | difference between | and nine The meaning is the same. ✓	

Models



Model 1: Interpret a Word Problem Using Only One Variable

The Fredricks used 12 more gallons (gal) of fuel oil in October than in September, and twice as much oil in November as in September. Write an expression for total gallons of fuel used in the three months, using only one variable.

Step 1	**Read and restate**	Determining the total number of gallons of fuel for three months
Step 2	**Information**	1. September is the first month. 2. October = (September) + 12 gal 3. November = 2 × September 4. Total
Step 3	**Relevancy**	All are relevant.
Step 4	**Target solution**	An expression for total fuel oil used in 3 months
Step 5	**Translation**	September fuel = x (Let x = September fuel) ⟶ **Why do we do this?** October fuel = $x + 12$ November fuel = $2x$
Step 6	**Model**	Answer: $\boldsymbol{x + x + 12 + 2x}$ **Do not simplify!** **Why not?**
Step 7	**Validate**	

September	+	October	+	November	3 months ✓
x	+	$x + 12$	+	$2x$	an expression ✓
fuel used in September		*twelve more gallons than fuel used in September*		*twice as much fuel used as in September*	meaning is the same ✓

Why do we do this? Recall that one definition of *variable* (from Section 1.1) is, "a placeholder in an expression that can assume any chosen value." When we translate a word problem into symbolic notation, it is up to us to choose that value. We must decide what idea or concept represents the most basic unknown — the one unknown, that, if we were to know the value of it, we could then solve for all the other unknowns through the process of substitution. Once we have done that, we represent that concept with x (or any other variable). Any time we assign a variable for a concept or unknown quantity, we must always define the value we are assigning to x with a statement, "Let x = ..." This is known as *defining a variable*.

In the case of Model 1, the information about all three months was given in terms of September's fuel usage. If we knew how many gallons were used in September, we would then be able to determine the fuel used in October and November. So it makes the most sense to "Let x = September fuel."

Why not? While it is customary to simplify such expressions, the point of this section is to build mathematical models that express the same meaning as the word problems on which they are based. If we were to simplify the expression, we would have a very different model which would no longer reflect the meaning of the original problem, despite the fact that the unsimplified and simplified versions would have equivalent mathematical meaning.

Model 2: Interpreting a Numbers Problem

Write a mathematical expression for the sum of three times a number squared and the difference of the number and seven.

Step 1 — Read and restate

Underline to indicate the terms to add.

The **sum of** three times a number squared and the difference of the same number and seven.

Step 2 — Information

1. sum (of ___ and ___)
2. 3 times
3. a number squared } *first term*
4. difference
5. the same number } *second term*
6. seven

Step 3 — Relevancy

all are relevant

Step 4 — Target solution

a mathematical expression for a sum of two quantities

Step 5 — Translation

1. + (will be the operation)
2. 3•
3. x^2 (a number, call it x, squared)
4. – (subtraction sign)
5. x
6. 7

Step 6 — Model

Answer: $\mathbf{3x^2 + (x - 7)}$

Step 7 — Validate

$3x^2$	+	$(x - 7)$	an expression ✓
three times a number squared	*the sum of*	*difference of the number and seven*	meaning is the same ✓

Model 3: Interpreting a Consecutive Number Problem in One Variable

Given two consecutive odd integers, write an expression for the sum of twice the first integer decreased by eleven and the second integer.

Step 1 **Read and restate** Write the sum of two times the first decreased by eleven and the second

Step 2 **Information**
1. first odd integer
2. second odd integer
3. sum (of ___ and ___)
4. twice
5. first odd integer
6. decreased by
7. 11
8. and
9. second odd integer

Step 3 **Relevancy** all are relevant

Step 4 **Target solution** an expression using two consecutive odd integers

Step 5 **Translation**
1. x (Let x = the first odd integer)
2. $x + 2$ (odd integers are 2 units apart)
3. + will be the operation
4. 2•
5. x
6. – (subtraction sign) } *first addend*
7. 11
8. $x + 2$ } *second addend*

Step 6 **Model** Answer: **For x = an odd integer: $(2x - 11) + (x + 2)$**

Step 7 **Validate**

$(2x - 11)$	+	$(x + 2)$	an expression ✓
twice the first odd integer decreased by eleven	*the sum of*	*the next consecutive odd integer*	meaning is the same ✓

Addressing Common Errors



Issue	Incorrect Process	Resolution	Correct Process
Using the wrong key word for the problem	Translate to symbolic representation: One half of thirty. $30 \div \frac{1}{2}$	Sometimes intuition interferes with math reasoning. Read closely and carefully. Use the table of key words to check that you are not misunderstanding what words or phrases translate to what symbols or operations.	$\frac{1}{2} \cdot 30$ or $\frac{30}{2}$ or $30 \div 2$ Key words are "half" and "of." "Half" indicates divide by 2, not divide by ½. Also "of" indicates **multiply** by ½, ***not divide*** by ½.

Validation

$\frac{1}{2} \cdot 30$	symbolic representation ✓
one half of thirty	meaning is the same ✓

Issue	Incorrect Process	Resolution	Correct Process
Incorrect word order with subtraction	Translate to symbolic representation: Eight less than three times a number $8 - 3x$	The key words for subtraction indicate in which order the phrase should be written. Validate that you have the numbers in the correct orientation by translating it back.	$3x - 8$

Validation

$3x$	-8	symbolic representation ✓
three times a number	*eight less than*	meaning is the same ✓

<table>
<tr><th>Issue</th><th>Incorrect Process</th><th>Resolution</th><th>Correct Process</th></tr>
<tr><td rowspan="3">Not validating by translating from symbols back to words</td><td>Translate to symbolic representation:
twice the sum of 11 and 4.
2 • 11 + 4</td><td>Underline the key words that indicate a phrase.</td><td>twice the sum of 11 and 4
twice means two times
2(11 + 4)</td></tr>
<tr><td colspan="3">Validation</td></tr>
<tr><td colspan="3"><table><tr><td>2 •</td><td>(11 + 4)</td><td>symbolic representation ✓</td></tr><tr><td>twice</td><td>the sum of 11 and 4</td><td>meaning is the same ✓</td></tr></table></td></tr>
<tr><th>Issue</th><th>Incorrect Process</th><th>Resolution</th><th>Correct Process</th></tr>
<tr><td rowspan="3">Missing the relevance of a word or phrase</td><td>Write in symbolic representation:
Jan worked 4 hours MWF and 6 on TTh. What is her pay if she makes $8 per hour?
4 • $8 + 6 • $8</td><td>Practical experience, common sense, real world knowledge are all a part of problem solving. Step 3 in the methodology is relevancy.</td><td>What do MWF and TTh mean to the problem?
MWF means 3 days, or
3 • (the number of MWF hours)
TTH means 2 days, or
(2 • the number of TTH hours)
$8 per hour means
$8 • the number of hours
$8[(3 • 4) + (2 • 6)]</td></tr>
<tr><td colspan="3">Validation</td></tr>
<tr><td colspan="3"><table><tr><td>$8 •</td><td>[(3 • 4)</td><td>+</td><td>(2 • 6)]</td><td>symbolic representation ✓</td></tr><tr><td>$8 pay per hour
“per” means “for each”</td><td>4 hours on 3 days (MWF)</td><td>and</td><td>6 hours on two days (TTH)</td><td>meaning is the same ✓</td></tr></table></td></tr>
</table>

<table>
<tr><th>Issue</th><th>Incorrect Process</th><th>Resolution</th><th>Correct Process</th></tr>
<tr><td rowspan="3">Missing the importance of punctuation when translating to symbolic representation</td><td>Translate to symbolic representation:
seven times some number, plus nine
$7(x + 9)$</td><td>Looking at key words alone does not always tell the whole story—you must also take punctuation (especially commas and periods) into account. Read carefully!</td><td>The comma helps us to determine where to place the parentheses:
(seven times some number), (plus nine)
$7x + 9$</td></tr>
<tr><td colspan="3">Validation</td></tr>
<tr><td colspan="3"><table>
<tr><td>(7 • x) + 9</td><td>symbolic representation ✓</td></tr>
<tr><td>seven times some number , plus nine</td><td>meaning is the same ✓</td></tr>
</table></td></tr>
</table>

Preparation Inventory



Before proceeding, you should be able to:

- ☐ Use the translation table to find the symbol for a given key word
- ☐ Use each step of the methodology for basic translation problems

Section 1.5

ACTIVITY

Interpreting Word Problems

Performance Criteria



- Translate a problem given in English to math symbols
  - key words identified
  - meaning clearly stated
  - appropriate interpretation
- Validate by translating from math symbols back into an equivalent English phrase
  - correct interpretation of symbols
  - correctly translated symbol to key word

Critical Thinking Questions



1. How do you determine what to include in a list of key words for a word problem?

2. How do you inventory key words?

3. How do you determine if there is a variable in a problem?

4. Why does the variable need to be defined?

5. How do you determine what is relevant in a problem?

6 What are possible types of word problems? You may include the types introduced in the models, but can you think of any additional types?

7. If you were translating a word problem and encountered the word “nearly” or “almost,” how might you handle translating it into symbolic representation? Explain.

Tips for Success



- Underline or highlight key words and phrases

Demonstrate Your Understanding



1. Translate the following expressions into math symbols.

Problem	Symbolic Phrase	Validation
a) The sum of a number and seven		
b) Three times the product of a number and five		
c) three times a number, increased by nine		
d) four times a number, minus ten		
e) The quotient of a number and nine		

Problem	Symbolic Phrase	Validation
f) six divided by one half		
g) six divided into thirds		

2. Write an expression using one variable to show the relationship between the numbers. Define the variable you choose.

Problem	Symbolic Expression	Validation
a) The sum of two consecutive integers		
b) Twice an odd integer subtracted from three times the next odd integer		
c) 4 times an integer added to 2 more than 3 times the next consecutive even integer		

3. Write a single mathematical expression, using only one variable, to show the relationship(s). Define the variable you choose.

Problem	Symbolic Expression	Validation
a) The sum of two numbers is 45		
b) A freezer costs $50 less than a refrigerator		
c) Nancy earns $7,000 less than twice what Kira earns.		

Problem	Symbolic Expression	Validation
d) On his business trip in New York, Rich's expenses for food and lodging combined were \$600 less than three times the cost of his airfare. (Hint: write an expression for the expense of food and lodging in terms of airfare.)		
e) Wendy is half as old as her professor, Dr. Rodriguez. Tom is four years older than Wendy. (Hint: write an expression for Tom's age in terms of Dr. Rodriguez's age.)		

4. Set up an expression to show how to find the quantity that is in italics. You may use more than one variable. Define each variable you use.

Problem	Symbolic Expression	Validation
a) The *amount of interest* is found by multiplying the amount of money (principle) times the interest rate (r) times the time (t).		
b) Your *intelligence quotient* is found by dividing your mental age by your chronological age and multiplying by 100.		
c) The *cost* of running your hairdryer is found by multiplying the wattage times the rate per kilowatt-hour times the time, in hours, all divided by 1000. 		

Problem	Symbolic Expression	Validation
d) You can determine the ***gas mileage*** of your car by dividing the distance traveled by the number of gallons of gas used.		
e) ***Earnings*** in tips are determined by the difference between the total amount collected and the cost of the items (selling price plus taxes).		

Identify and Correct the Errors



In the second column, identify the error(s) in the worked solution or validate its answer.
If the worked solution is incorrect, solve the problem correctly in the third column and validate your answer.

Worked Solution	Identify Errors or Validate	Correct Process	Validation
1) Translate to symbols: Four times an odd integer is added to half the next consecutive odd integer $4(x+1)+\frac{1}{2}(x+2)$			
2) The price of a share of stock rose \$1.20. Write an expression showing the price of 15 shares. $(x+\$1.20)15$			
3) Translate: Eight is added to the difference of a number and three times the number. $8+(3x-x)$			
5) Translate into symbolic representation: one-third of a number added to six minus 18 $\left(\frac{1}{3}x+6\right)-18$			